Who is on the other side of your trade?

How exchange rules distribute trading costs and risks, and what this means for protecting everyday traders

Rent or toxicity: strategic exploitability of learning market makers with public wallet information

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6.S974 Games, Learning, and Security, Workshop 1
Example 4: an algorithmic market with adaptive agents

Good morning. We are Deepu and Miguel, Example 4: an algorithmic market with adaptive agents. Our long-term motivation is to help everyday traders avoid preventable losses. On Hyperliquid, every fill carries public wallet addresses, which lets us study how information and exchange rules distribute trading costs and risks. Today we establish mechanisms: how learned maker profits can be attacked, when identity helps, and who bears toxic flow in our model. We have not yet demonstrated a rule that protects retail accounts from severe losses. That is the next research stage, with customer outcomes as the test of success. Ten minutes: two on why, three on the model, four on evidence, one on that next step.

2026: the year retail leverage went wrong at national scale

−44%
KOSPI, June 19 to July 29

1.2 million
retail accounts margin-called:
3.4% of all adults

$39 billion
retail losses on single-stock leveraged ETFs

"Retail" means everyday people

- 92% of the holders were individuals, not firms
- 62% of the accounts: people aged 20 to 39
- taking leveraged exposure through products whose risks can be hard to assess

The public-benefit goal

Help people avoid preventable trading losses.

Today: identify mechanisms that distribute trading costs and risks.

Next: test whether changing the rules improves ordinary traders' outcomes.

Sources for background figures: Bloomberg, Reuters, Goldman Sachs, Citi (via Seoulz, "KOSPI Crash 2026"); Maeil Shinmun, Oct 22, 2026.

Why we care: large trading losses affect people with ordinary jobs, savings, and families. The Korean episode is our motivation, not an outcome our Hyperliquid model explains. We cannot attribute those losses to the mechanisms studied here. Our present contribution is narrower: identify how information and exchange rules distribute execution costs and risks. The public-benefit goal is to use those mechanisms to test protections, and to judge them by whether less-informed traders are better off. Account liquidation requires a further model extension.

Leverage magnifies losses; liquidation is a future extension

You put in $1,000. At 20× leverage you control a $20,000 position.



collateral: $1,000 **exposure beyond collateral**: $19,000

price +5%

position gain: $20,000 × 5%

you doubled your money: +$1,000

price −5%

position loss: $20,000 × 5%

loss equals collateral: −$1,000

illustrative PnL before costs; maintenance margin can trigger liquidation before this point

Current model: quotes, fills, information and queue position. It does not track leveraged accounts.

Later extension: collateral, margin and forced sales, to test protections against liquidation losses.

Leverage means exposure larger than the collateral supporting it. A thousand dollars supporting a twenty-thousand-dollar long position gives twenty-times leverage. A five-percent rise produces a thousand-dollar gain before fees and funding; a five-percent fall produces a loss equal to the original collateral. For perpetuals this is exposure supported by margin, rather than a literal nineteen-thousand-dollar loan from the exchange. Maintenance margin can trigger liquidation before equity reaches zero, and liquidation can leave collateral remaining. Leveraged ETFs have a different structure, typically with daily resetting; the Korean products and a perpetual account should not be treated as the same liquidation mechanism. Our current simulator studies execution and information. Modelling collateral, margin, depth and forced sales is a later extension needed before we can claim to reduce liquidation losses. Sources for these distinctions: Hyperliquid Docs, Margining and Liquidations; SEC Investor Bulletin, Leveraged and Inverse ETFs.

From trading mechanisms to protections we can test

What this study establishes

- learned maker profits can be vulnerable to a strategic entrant
- wallet information can help select fills in the model
- identity and priority rules change who bears toxic flow

The missing test

Does changing a rule improve outcomes for traders with less information and fewer resources?

Public wallets make the question measurable

fill buy 0.5 BTC @ 67,412.0
taker `0x7f3a...c91e` maker `0x21b0...4d7a`
a wallet's history is observable

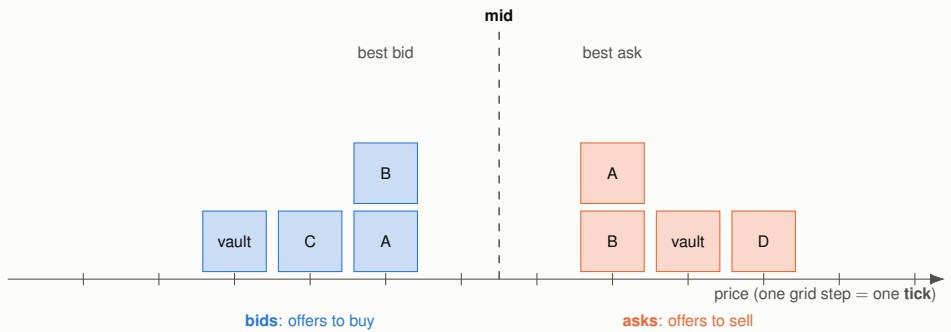
Next: measure customer outcomes

- execution costs and gains from trade
- access to liquidity during stress
- severe losses for passive traders
- whether improvements survive adaptation

Consumer protection is the next test. Current evidence identifies mechanisms and tradeoffs.

Here is the bridge from the human motivation to what we can actually establish. Today we show strategic vulnerabilities and how information and priority redistribute trading outcomes. An attacker earning more does not establish that customers benefit; it can simply move profit between sophisticated firms. Our next test compares exchange rules by less-informed traders' costs, gains from trade, access in stress, and severe losses. We will separate small takers from passive liquidity providers, because the same rule can help one and hurt the other. Public wallet histories let us measure behavior, but a wallet does not tell us whether its owner is retail or wealthy. The current work motivates protective interventions; their effectiveness remains to be tested.

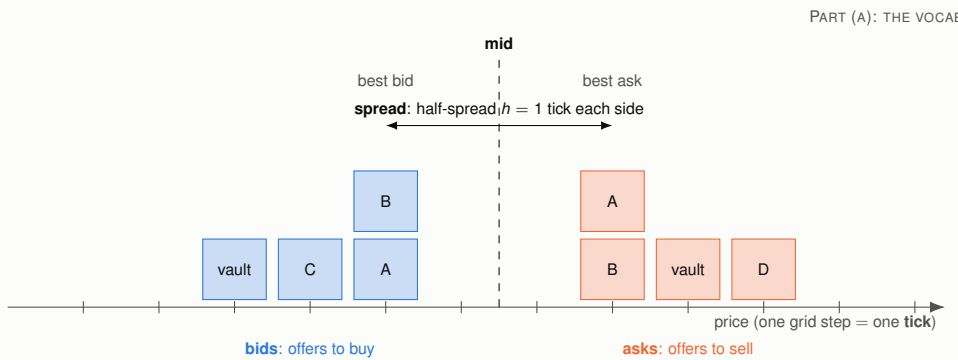
An exchange is a list of standing offers: the order book



Makers post the offers (the boxes). **Takers** accept them.
Each box carries a public **wallet** address.

Vocabulary, because the audience has not seen this before. An exchange is a list of standing offers, the order book. Price lives on a grid; one grid step is a tick. On the left, in blue, are bids: offers to buy at that price. On the right, in orange, are asks: offers to sell. The people who post these standing offers are market makers; each box is one maker's order and on Hyperliquid each box carries the maker's public wallet address. The dashed line is the mid, halfway between the best bid and the best ask. [click] The distance between best bid and best ask is the spread; we write the half-spread h in ticks, here one tick each side. [click] A taker is someone who accepts an offer instead of posting one. T buys one unit: he hits the best ask, and that is a fill for maker B, because B was first in line at that price and A was behind. A bigger order takes A's too and walks to the next price, and that walking is most of what a trade really costs. What becomes public: the fill itself, buy, price, both wallets, immediately. What does not: whether B made or lost money on it. That only shows up as the price moves afterwards, which is the next slide.

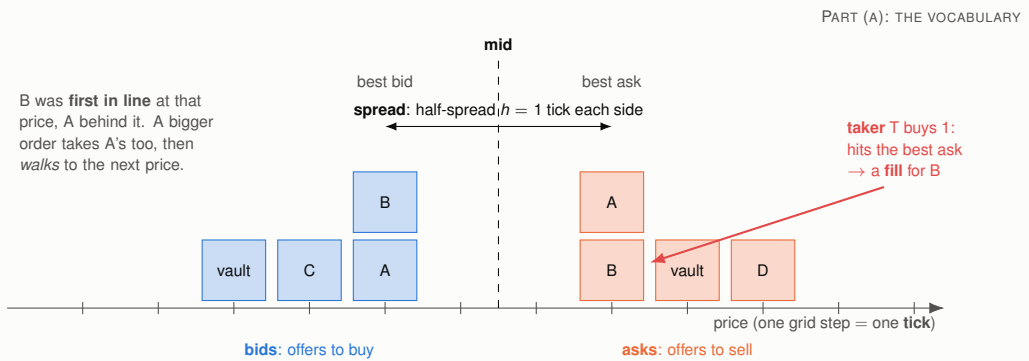
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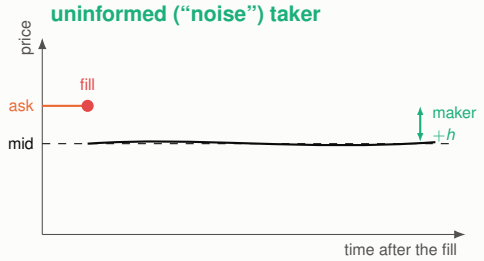


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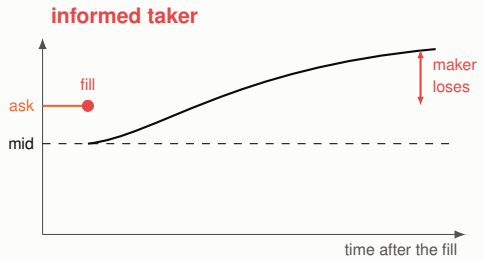
Public, at once: buy, price, wallet T, wallet B.
Public, only later: whether B gained or lost on it.

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Did the maker win? Look at the price afterwards: the markout



price does not move; the maker keeps the half-spread



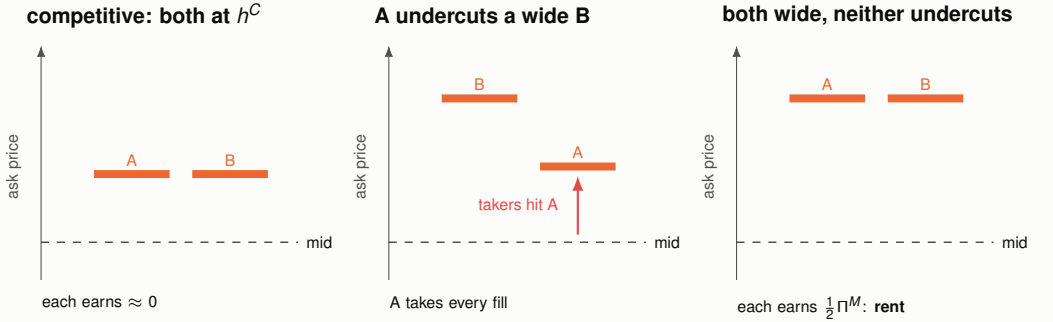
the taker knew; the price runs past the fill price

markout = mid a few seconds later − fill price, signed for the maker

toxic flow: a stream of fills with negative markouts (*adverse selection*)

How do we know whether the maker won on a fill? We look at where the price goes afterwards. On the left, the taker was uninformed, a noise trader; the price sits where it was, and the maker sold above the mid, so it keeps the half-spread. On the right the taker knew something; the price runs up past the price the maker sold at, and the maker has sold too cheap. That signed difference, mid a few seconds later minus the fill price from the maker's point of view, is the markout. It is the single measurement everything in our data work rests on. A stream of fills with negative markouts is what practitioners call toxic flow, and economists call adverse selection. A maker's whole business is being in line for the left-hand fills and not the right-hand ones.

Rent: what makers earn by not undercutting each other

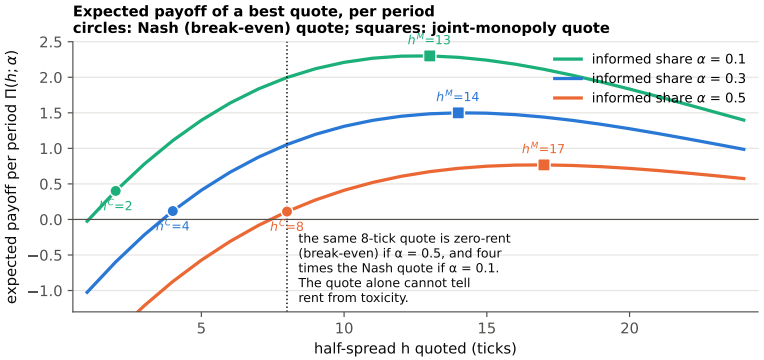


h^C: the **competitive** quote, break-even. Wider pays, but invites undercutting.

Learning makers end up in the right panel *without talking*: **tacit collusion**. The extra is the **rent**.

The other word in the title: rent. Three pictures of the same two makers' ask prices. Left: both quote the competitive half-spread h-C, where what you earn from noise takers just covers what you lose to informed ones; each earns about zero. Middle: B quotes wide and A undercuts by a tick; takers always hit the tighter price, so A takes every fill and B gets nothing. That is why a wide quote is fragile. Right: both quote wide and neither undercuts. Each takes half the fills at the wide price and earns half the monopoly profit. The difference between the right panel and the left is the rent. The striking fact, from Colliard, Foucault and Lovo and replicated by us, is that independent learning algorithms end up in the right panel without ever communicating. Each learns that undercutting today gets punished tomorrow, and stops. That is tacit collusion, and the rent is what the uninformed taker pays for it.

The puzzle: on the tape, rent and toxicity look identical

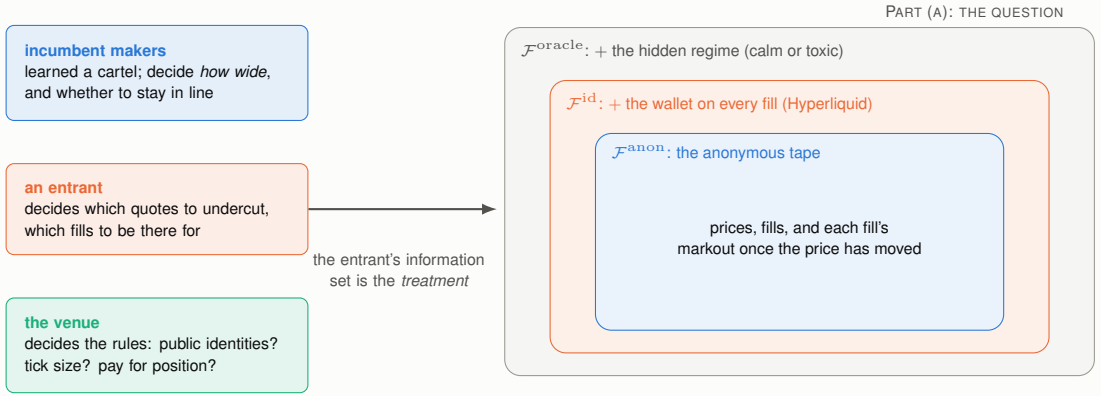


A wide quote is a **rent** . . . or the correct charge for **toxic flow**.

Which curve you are on is decided by α, the share of informed takers. Only outcomes, or *who* trades, reveal it.

Here is the puzzle. This is the maker's expected profit per period as a function of its half-spread, one curve per level of toxicity alpha, the fraction of takers who are informed. Each curve has a break-even point on the left, h-C, and a peak, h-M. More toxic flow pushes the whole curve down and both points to the right. So if you watch a market and see wide quotes, you cannot tell whether the makers are colluding on a calm market, the blue curve, or correctly charging for a toxic one, the red curve. Rent and toxicity produce the same tape. The only ways to tell are to look at outcomes, the markouts, or to look at who is trading. And that second route exists only because wallets are public.

Three players, and what each one gets to know



Who plays. Three players. The incumbents, in blue, are the makers who have already learned a cartel; they decide how wide to quote and, on a one-tick book, whether to stay in the queue. The entrant, in orange, is the adversary: a new maker who decides which quotes to undercut and which fills to be there for. The venue, in green, sets the rules: are identities public, how coarse is the tick, can a maker pay for queue position. The right side is the information structure, nested. The anonymous tape, which every exchange in the world shows: prices, fills, and eventually each fill's markout. Add the wallet on every fill and you have what Hyperliquid shows. Add the hidden regime, whether the market is currently calm or toxic, and you have an oracle that we use only as an upper bound. The entrant's information set is the treatment: we run the same attacker with each of the three and the difference is what identity is worth.

What we want to learn

RQ1 Value of identity

What is a public wallet worth to an entrant, beyond the tape, and through which mechanism?

RQ2 Exploitability

How much of a learned cartel's rent can an adversary take or destroy?

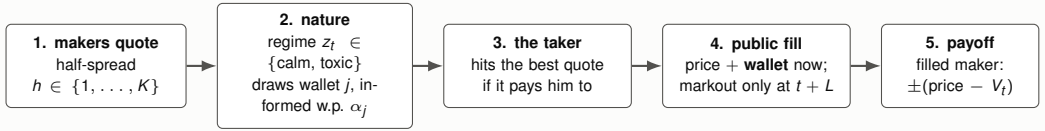
RQ3 One-tick books

When the grid pins the spread, where does rent live, and which rules decide who holds the toxic flow?

Today: identify mechanisms. Next: test rules by customer outcomes.

Our three current questions concern identity, exploitability and venue rules. They explain how strategic advantages arise and who bears toxic flow in the model. The players adapt to each other, so changing a rule can change both quoting behavior and the outcomes customers receive. Our next-stage question is whether an intervention lowers less-informed traders' costs and severe losses while preserving access in stress. That benefit must be measured separately from the attacker's profit or the makers' resilience.

One period of the game



Incumbents: tabular Q-learners, state = (own last h, rival's last h); trained, then **frozen**.

Entrant: information set $\mathcal{F}^{\text{anon}} \subset \mathcal{F}^{\text{id}} \subset \mathcal{F}^{\text{oracle}}$; a Bayesian filter estimates the next taker's α.

Payoff curve, closed form:
$\Pi(h;\alpha) = (1-\alpha)\,2x\bar{\Phi}(x/\sigma_L) - \alpha\mu e^{-x/\mu}$,
$x = h\theta$.

The model, one period. Step one: every maker commits a half-spread h on the tick grid. Step two: nature draws a regime, calm or toxic, from a persistent Markov chain, then a wallet from a population of fifty; each wallet has its own probability alpha-j of being informed, and in the toxic state the active wallets are the toxic ones. Step three: the taker hits the best quote if it pays him to, which for an informed taker means exactly when the maker will lose. Step four: the fill is public at once, price and wallet, but its markout, the value V-t the maker traded against, is only revealed L arrivals later. That lag L is the key parameter. Step five: the filled maker's payoff is price minus value. One hundred periods make an episode; the regime chain and the wallets persist across episodes. The incumbents are tabular Q-learners whose state is their own last quote and the best rival's; they are identity-blind by construction; we train them twenty thousand episodes and then freeze them, so a cartel is a fixed reaction function. The entrant's information set is the treatment, and a Bayesian filter turns whatever it sees into an estimate of the next taker's alpha. And because the market is Glosten-Milgrom, the expected payoff of a quote against informed share alpha is closed form, which is what makes rent measurable rather than estimated.

Benchmarks are closed-form, so “rent” is measured, not estimated

$h^C = \min\{h : \Pi(h) \geq 0\}$	**Nash** (competitive)
$h^M = \arg\max_h \Pi(h)$	**joint monopoly**
$C = \dfrac{\Pi_{\text{cartel}} - \Pi^C}{\Pi^M - \Pi^C} \in [0,1]$	**rent index**
$L_{\text{BR}} = \Pi_{\text{cartel}} - \Pi_{\text{under attack}}$	**loss under attack**
$\kappa_A = L_{\text{BR}}/(\Pi_{\text{cartel}} - \Pi^C)$	**rent destroyed**

At $\bar{\alpha} = 0.3$: $h^C = 4$, $h^M = 14$, rent $\Pi^M - \Pi^C = 1.38$.

Why a cartel is attackable: it is a *reaction function* that does not chase every undercut. That restraint is the rent, and the attack surface.

One-tick (queue) mode: spread pinned; actions REST / OUT / REST-AND-PAY, first in first out. A fill at h^C: $+1.10$ calm, -2.26 toxic.

Because the payoff curve is closed form, the benchmarks are exact. h-C is the smallest break-even quote, the Nash outcome of one-shot competition: any wider quote is profitably undercut. h-M is the joint-monopoly quote. The rent index C puts a frozen cartel between zero, Nash, and one, monopoly; the cartels we train sit at 0.7 to 0.9. Under an adversary A, L-B-R is what the cartel loses, and kappa-A is that loss as a share of the rent it had; kappa above one means the attacker destroyed more than the rent. All our results are in rent units, divided by Pi-M minus Pi-C, so cartels at different toxicity levels are comparable. A worked example at alpha 0.3: two incumbents at fourteen ticks split 1.50 per period. An entrant at thirteen takes every fill and earns 1.49, they earn zero. If they chase, the entrant goes to twelve, and the chase ends at h-C equals four, where everyone earns 0.12. A learned cartel is a reaction function that does not chase every time; that restraint is the rent, and it is also the attack surface, because the attacker can learn exactly when it chases and when it does not. The queue mode models the one-tick book: the spread is pinned, so the only decisions are rest, step out, or pay for priority, and the fill queue is first in first out. At break-even a fill is worth plus 1.10 in a calm spell and minus 2.26 in a toxic one, so all the value is in which fills you take.

The prototype, in one command

`python experiments/demo.py`

1. benchmarks h^C, h^M, $\Pi(h)$
2. the shipped cartel alone: $C = 0.86$
3. three adversaries enter
4. the best response, period by period

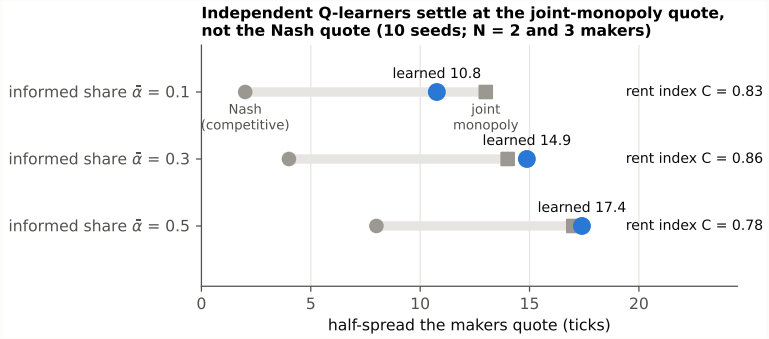
`github.com/pradeepu191/wargame`
59 tests; every result logged with its command.

```
Entry (200 episodes each, same random numbers):
  adversary                        cartel loses  entrant earns  kappa_A
  undercut (one tick inside)              0.41           0.33     0.48
  markout (undercut if rent inferred)     0.47           0.41     0.55
  exact best response                     0.94           1.09     1.10

Trace: exact best response vs the cartel (quotes in ticks)
   t  inc0  inc1  entrant   taker         filled   payoff
   0    15    24       12   noise    buy  entrant   +1.21
   1    23    16       13   noise    buy  entrant   +7.86
   2    20    14       13   no trade
   5    21    19       14   informed buy  entrant   -5.77
   7    18    23       14   noise    buy  entrant  +11.79
```


[0:15] Everything you are about to see runs from one command in seconds: benchmarks, the cartel alone, three adversaries entering under the same random numbers, and the best response traced period by period. Fifty-nine tests, every number in the report logged with the command that produced it.

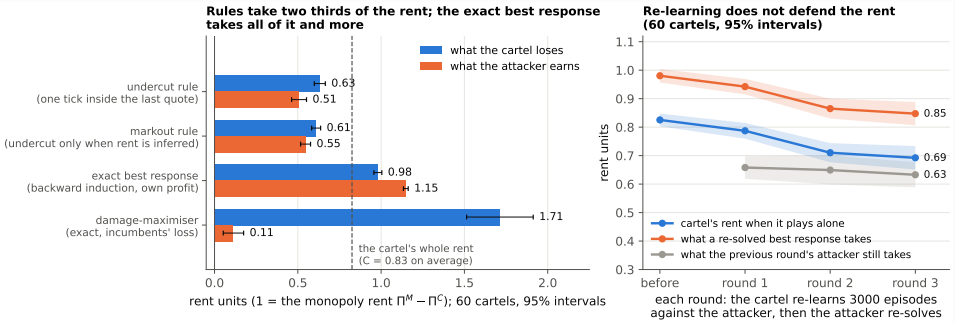
Evidence 1: learners coordinate above the Nash point



$C = 0.71$ to 0.87 in every cell. In 55 to 83% of states undercutting would pay *this period*; the learner forgoes it 96 to 100% of the time.

[0:30] First, the thing we attack exists. Independent Q-learners settle at a rent index of 0.7 to 0.9 in every cell of a grid over the number of makers, learning rate and initialisation; that replicates Colliard, Foucault and Lovo. And we can see the mechanism: in most states undercutting would pay this period, and the learner almost never does it. It has learned that the rival punishes. That restraint is the rent, and it is also the attack surface.

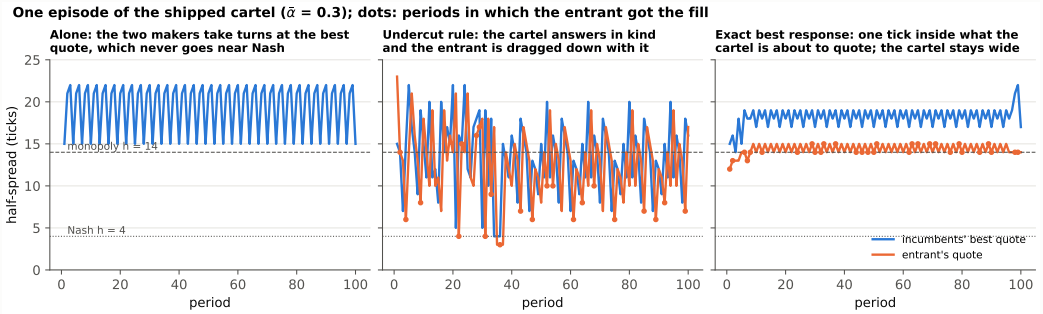
Evidence 2: the rent is fully exploitable



Hand-written rules take two thirds. The **exact best response** takes it all, in 60 of 60 cartels: $\kappa_A = 1.1$ to 1.3.
The damage-maximiser destroys 2 to 3$\times$ the rent at no cost.

[0:45] Two hand-written rules take about two thirds of the rent. The exact best response solves the frozen cartel's reaction function by backward induction and removes its entire rent in sixty of sixty tested populations, earning monopoly-like profit for itself. Re-solving after the cartel re-learns preserves the attack. This establishes strategic vulnerability under the model's assumptions. It does not show that customers receive cheaper execution: the entrant can keep the market wide and collect the profit itself. For our next stage, this attacker is a stress-testing instrument. The separate question is whether a protective rule improves customer outcomes under that stress.

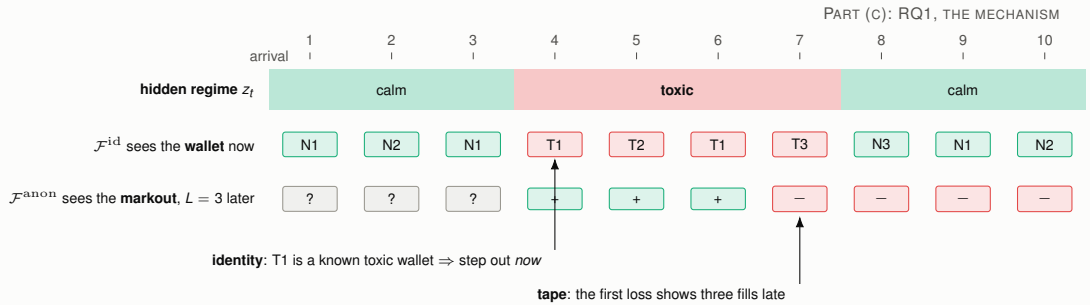
Steering, period by period



Left: the cartel alone rotates at wide quotes. **Middle**: a naive undercut starts a price war. **Right**: the best response holds the cartel wide and takes the fills.

[0:30] What that looks like. Left: the cartel alone rotates at wide quotes and never goes near Nash. Middle: a naive undercut starts a price war and both sides slide to break-even. Right: the best response never provokes the war. It quotes one tick inside the reaction it has predicted; the cartel's learned answer is to stay wide; and the entrant takes nearly every fill at the monopoly price. The cartel's own restraint does the work.

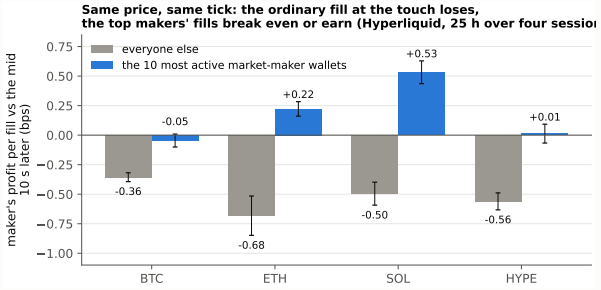
Why identity has value: a spell of toxic flow, seen two ways



Identity is worth what you lose in the L fills before the tape catches up: measured, zero at $\bar{\alpha} \leq 0.3$, growing with L at $\bar{\alpha} = 0.5$ (backup).

[0:40] Now RQ1, what a wallet is worth beyond the tape. Informed trading comes in spells, and the toxic wallets are the ones active in a spell. The regime turns toxic at arrival four. With identities, the fourth fill carries a wallet you have seen lose you money before, so you step out now. The anonymous tape learns a fill's markout L arrivals later; with L equal to three, the first minus sign appears at arrival seven. So identity is worth what you lose on those L fills: measured, it is zero where even toxic flow leaves an undercut profitable, alpha up to 0.3, and grows with the lag at alpha 0.5, up to the lifetime of a spell. On Hyperliquid that lag is real: the outcome of a fill is knowable only once the price has moved; the wallet is on the fill at once.

Evidence 3, Hyperliquid: the price cannot move, and someone still earns



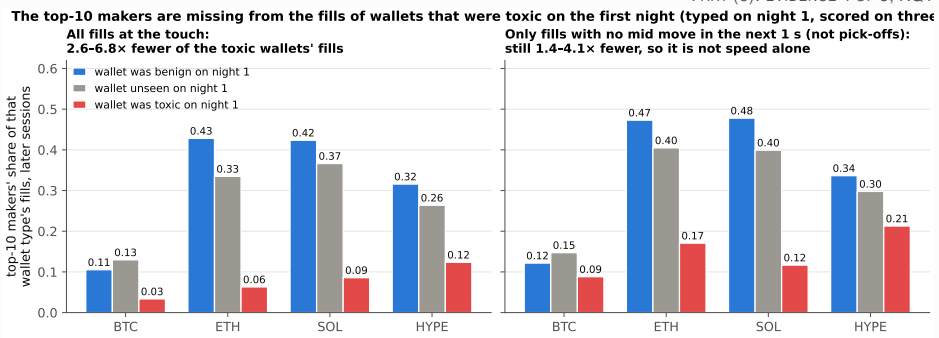
Four sessions, 25 h, 490k scored fills.

- **BTC/ETH/SOL spreads**: essentially one tick
- **Average maker markout**: negative at 10 s on every coin
- **The top makers earn** $+0.3$ to $+1.0$ bps more, *at the same price*; rank persists (corr. 0.5 to 0.8)

Fill selection matters even at the same quoted price.

[0:35] Four sessions, twenty-five hours, four hundred and ninety thousand scored fills. BTC, ETH and SOL spreads are essentially one tick. Average maker markouts at the touch are negative at ten seconds, while the top ten active makers do better at the same price. These are outcomes for the maker on each fill, not retail account returns. The comparison shows that fill selection matters; it does not by itself establish excess profits after all operating costs. For consumer protection, the next step is to measure outcomes separately for customers taking liquidity and customers supplying it.

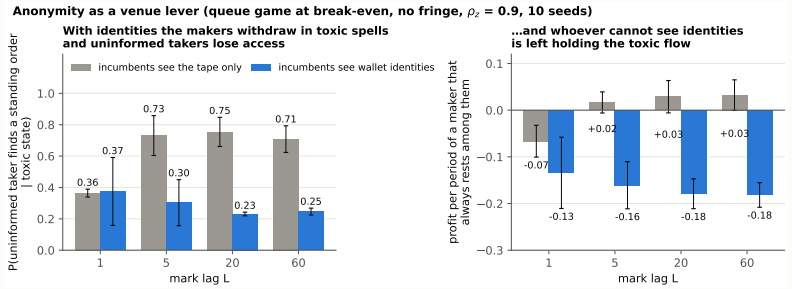
Evidence 4: top makers receive fewer toxic-wallet fills



Wallets typed on night 1, scored on three later sessions. Consistent with selection; the makers' decision signals remain unobserved.

[0:35] We typed taker wallets on the first night and measured the top makers' share of their fills in later sessions. Their share is lower for previously toxic wallets, in calm and stressed tape, and the gap survives when the mid does not move within one second. This is evidence of differential exposure consistent with selection. It does not identify the signals the makers actually used or prove that wallet identity caused the gap. The data motivates our information treatments and a closer study of cancellations and queue actions. We also cannot label these wallets as ordinary households from their addresses alone.

Evidence 5: venue rules redistribute access and losses



Anonymity: more uninformed takers get filled in stress in the queue model.

Priority fee: venue revenue rises; the always-resting fringe's losses deepen.

Next test: customer costs and severe losses, alongside access and maker viability.

[0:30] This is our closest current result to the consumer-protection question. In the queue model, anonymity raises uninformed takers' fill probability during toxic spells. Priority fees raise venue revenue and deepen the losses of the always-resting fringe. These are distributional tradeoffs. More fills alone do not establish higher welfare, and the fringe is a modeled strategy rather than an identified retail population. The next test measures costs, gains from trade and severe losses separately for takers and passive traders, while checking that makers still supply liquidity.

Workshop 2: test protections for less-informed traders

1 Customer outcomes

Separate small **takers** and **passive traders**: execution costs, gains from trade, stress access and worst-5% losses.

2 Compare rules

Public, anonymous or delayed wallet disclosure. FIFO versus paid priority. Measure **who gains and who loses**.

3 Stress-test benefits

Let makers adapt. Use the existing attacker as a stress test. Check liquidity and whether loss-making participants leave.

Your view: which rule improves less-informed traders' outcomes, and does the improvement survive stress and adaptation?

[0:35] The priority for Workshop 2 is a consumer-protection experiment. First, define small takers and passive traders separately and measure execution costs, gains from trade, access in stress and worst-five-percent losses. Our existing private valuations can support a model-based gains-from-trade measure. Second, compare identity disclosure and priority rules under matched market conditions and multiple seeds. Third, let makers adapt and use the existing attacker to test whether benefits survive. We must also let participants leave if they keep losing, rather than assume permanent subsidizing liquidity. The test of success is better customer outcomes with viable liquidity. Our question for the room is which rule improves those outcomes, and whether that improvement survives stress and adaptation.

Longer term: connect execution risk to liquidation losses

PART (D): FUTURE EXTENSION, NOT A CURRENT RESULT

Add to the model

- persistent prices and finite book depth
- account collateral and maintenance margin
- forced sales and their price impact

Compare: leverage limits, margin buffers and liquidation execution rules.

Measure protection directly

- fraction of small accounts liquidated
- collateral remaining after liquidation
- severe losses and ability to exit in stress

Evidence: longer recording windows and stress episodes to calibrate the extension.

> **Reducing liquidation losses is a future hypothesis. Today's model cannot establish it.**

[0:25] This longer-term extension connects most directly to the people whose losses motivated us. We would add persistent prices, finite depth, collateral, maintenance margin, and forced sales with price impact. Then compare candidate protections by the fraction of small accounts liquidated, collateral retained and severe losses, while checking their ability to exit. Leverage limits, margin buffers and liquidation rules are hypotheses to test. The present model cannot establish that any of them prevents account wipeouts. Longer recording windows and stress episodes will help calibrate this extension.

If you remember three things

1 Vulnerabilities

Learned maker profits are exploitable in our model. This gives us a way to stress-test market rules.

2 Distribution

Information and priority change who gets fills and who bears losses. The data motivates testing these mechanisms.

3 Protection next

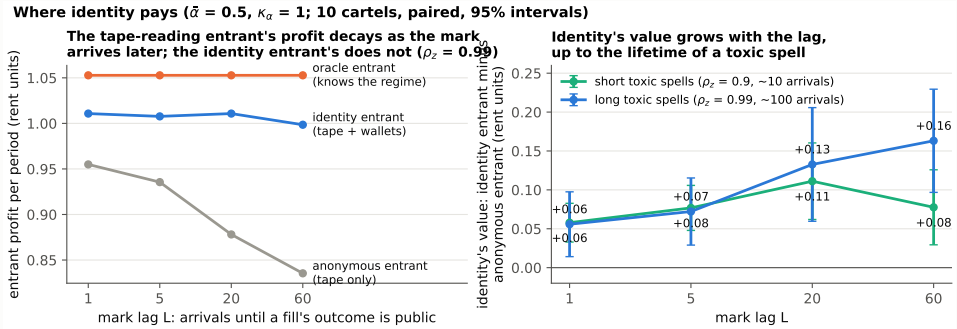
Test rules by customer costs, access and severe losses. Measure whether less-informed traders benefit.

Goal: fewer preventable losses for everyday traders.

`github.com/pradeepu191/wargame`

[0:25] Three things to remember. We have established vulnerabilities in learned maker behavior. We have shown how information and priority redistribute outcomes in the model, with empirical patterns that motivate those mechanisms. Now we will judge interventions by outcomes for less-informed traders. Helping an attacker profit is not sufficient evidence of public benefit. The next contribution must show whether a rule lowers customer costs or severe losses while preserving access to liquidity. Reducing leveraged account losses requires the later liquidation extension. That is how we plan to turn this work toward fewer preventable losses for everyday traders. Thank you.

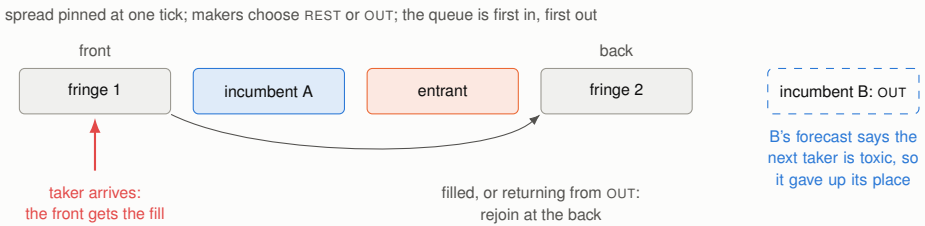
Backup: what identity is worth, measured



The proposal's mechanism (the same wallet trades again) is worth nothing: repeats are 2 to 5%. The lag mechanism survives: zero where $\bar{\alpha} \leq 0.3$, growing with L up to the spell's lifetime.

Measured. Our original proposal's mechanism, that the same wallet trades again so you can learn it, is worth nothing on the data: a wallet repeats within the window two to five percent of the time. The mechanism that survives is the lag story from the previous slide. The value of identity is zero wherever even the toxic state leaves an undercut profitable, alpha up to 0.3, because then you do not care who is coming. It becomes positive at alpha 0.5 and grows with the mark lag L, saturating at the spell's lifetime. So RQ1's answer is: identity is a lag story, and its value depends on the venue's reporting delay and the market's toxicity, not on recognising individuals.

Backup: the queue game, the one-tick book



A fill at h^C: $+1.10$ in a calm spell, -2.26 in a toxic one.

Who pays in the model: the always-resting *fringe*. Its retail interpretation needs testing.

The spread is fixed, and makers choose whether to rest or step out. Resting makers form a first-in-first-out queue. A filled maker, or one returning after stepping out, joins the back. Selection avoids toxic fills at the cost of queue position. The always-resting fringe absorbs flow that selectors avoid. That strategy may approximate some passive liquidity, but it is not an empirically identified retail group or a calibrated vault policy. The consumer-protection extension will give small passive traders explicit constraints and allow loss-making participants to leave. We also model paid priority, which changes who receives favorable fills.

Backup: Hyperliquid numbers

	BTC	ETH	SOL	HYPE
one tick (bps)	0.12	0.39	0.86	0.11
taker's cost vs the mid: fills at the touch / walked (bps)	0.12 / 2.35	0.42 / 3.20	0.92 / 5.20	0.17 / 3.57
10-s maker markout, at the touch / walked (bps)	-0.41 / -0.80	-0.56 / -2.26	-0.27 / $+0.98$	-0.45 / -0.01
next fill's markout after a toxic vs a benign wallet (bps, 327 h)	-0.64	-1.01	-0.59	-0.59
10-s markout at the touch: top-10 makers / everyone else (bps)	-0.05 / -0.36	$+0.22$ / -0.68	$+0.53$ / -0.50	$+0.01$ / -0.56
per-maker rank correlation, first session vs later (n makers)	0.80 (113)	0.53 (80)	0.70 (53)	0.59 (70)
top-10 share of touch fills: benign / unseen / toxic wallets	.11 / .13 / .03	.43 / .34 / .06	.42 / .37 / .09	.32 / .26 / .12
. . . on fills with no mid move in 1 s: benign / toxic	.12 / .09	.47 / .17	.48 / .12	.34 / .21

Four book sessions over two weeks (25 h). Standard errors cluster by five-minute block; every top-10-vs-rest gap exceeds four SEs. Wallet types from the first night, scored on the three later sessions.

Backup: the double oracle and robustness, in numbers

- **Double oracle** (cartel re-learns 3000 episodes against the fixed best response, entrant re-solves; 3 rounds): the now-stale attacker's profit falls 1.2 $\rightarrow$ 0.5 (the undercut-rule level); the re-solved one takes $\geq$ 90% of the remaining rent in every cartel every round; the cartel's own rent erodes 0.83 $\rightarrow$ 0.79 $\rightarrow$ 0.71 $\rightarrow$ 0.69.
- **Damage-maximiser** ($\lambda = 0$): steers the cartel to 4 to 7 ticks, inside break-even ($h^C = 8$ to 13), by quoting above it; takes no fills, earns ≈ 0; $L_{\mathrm{BR}} = 1.3$ ($\bar{\alpha} = 0.3$), 2.7 to 2.8 ($\bar{\alpha} = 0.5$).
- **Train at $\bar{\alpha}$, test at $\bar{\alpha}'$**: cartels keep $C = 0.80$ to 0.86, $\kappa_A = 0.65$ to 0.75 (rules) on calmer markets; on a more toxic one the rent collapses ($C = 0.23$) and is more than destroyed.
- **Held-out adversaries** preserve every defence ordering (rank correlation 0.75 to 0.96).
- **Caveat we own:** the exact best response needs the cartel's reaction function (oracle knowledge of the *policy*). It is the true upper bound; the public-information attacker that identifies it is next.

Backup: the queue game and the priority fee

- Learned incumbents need a **fringe** of always-resting makers (else a toxic spell empties the book for good); tabular Q cannot learn to queue (time priority pays only after consecutive periods in line); **no-regret (EXP3)** incumbents over complete policies learn to select in a few hundred episodes.
- Against selecting incumbents identity is worth *more*: $+0.05$ to $+0.06$ per period at every lag, 5/5 seeds; the fringe's payoff falls to -0.02 to -0.10: identity-blind resting liquidity is the counterparty of the selection rent.
- **Priority fee**: every selector pays in calm periods (identity-aware incumbents 93 to 94% of calm, 8 to 9% of toxic periods); a non-paying entrant among payers gets 0% of fills; the fringe's loss triples at any fee; revenue peaks at $f = 0.5$.
- **Makers' levers** (20 cartels, 7 adversaries incl. a Q-learning one): requoting every 2 to 5 periods costs everything; randomising costs profit and buys nothing; the identity floor: worst-case profit 0.08 $\rightarrow$ 0.19 at $\bar{\alpha} = 0.5$.